

Experiment 6 HDOC
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Batch - 15

Question 1

Write a C program to find the side (cm), perimeter (cm), and diagonal (cm) of a square when its area is given in square meters.

Step 1: Problem Understanding

Input

Area of square (in square meters)

Output

Side (cm)
Perimeter (cm)
Diagonal (cm)

Formula

$$A = s^2$$

$$\text{Side (m)} = \sqrt{\text{Area}}$$

$$\text{Side (cm)} = \text{Side (m)} \times 100$$

$$\text{Perimeter} = 4 \times \text{Side (cm)}$$

$$\text{Diagonal} = \text{Side (cm)} \times \sqrt{2}$$

Step 2: Test Case Table

Test Case	Input (Area m ²)	Expected Output	Actual Output	Result
1	1	Side=100 cm, Perimeter=400 cm, Diagonal=141.42 cm	Same	Pass
2	4	Side=200 cm, Perimeter=800 cm, Diagonal=282.84 cm	Same	Pass
3	9	Side=300 cm, Perimeter=1200 cm, Diagonal=424.26 cm	Same	Pass

Step 3: Algorithm

1. Start.
2. Read the area in square meters.
3. Calculate side in meters using square root.
4. Convert side into centimeters.
5. Calculate perimeter.
6. Calculate diagonal.
7. Display the results.
8. Stop.

C Program

```
#include <stdio.h>
#include <math.h>

int main()
{
    float area, side_m, side_cm, perimeter, diagonal;

    printf("Enter area of square (in square meters): ");
    scanf("%f", &area);

    side_m = sqrt(area);
    side_cm = side_m * 100;

    perimeter = 4 * side_cm;
    diagonal = side_cm * sqrt(2);

    printf("\nSide = %.2f cm", side_cm);
    printf("\nPerimeter = %.2f cm", perimeter);
    printf("\nDiagonal = %.2f cm", diagonal);

    return 0;
}
```

```
UM PICO 5.09 File: experiment6.c Modified
#include <stdio.h>
#include <math.h>

int main()
{
    float area, side_m, side_cm, perimeter, diagonal;

    printf("Enter area of square (in square meters): ");
    scanf("%f", &area);

    side_m = sqrt(area);
    side_cm = side_m * 100;

    perimeter = 4 * side_cm;
    diagonal = side_cm * sqrt(2);

    printf("\nSide = %.2f cm", side_cm);
    printf("\nPerimeter = %.2f cm", perimeter);
    printf("\nDiagonal = %.2f cm", diagonal);

    return 0;
}
```

Step 5: Evaluation

Compile and execute.

Step 6: OUTPUT

```
aaryanrajput@searyans-macbook ~ % ./experiment6.c
Enter area of square (in square meters): 200

Side = 1414.21 cm
Perimeter = 5656.85 cm
Diagonal = 2000.00 cm
aaryanrajput@searyans-macbook ~ %
```

Question 2

Write a C program to find the remaining area of a square sheet when a quadrant of given radius is removed from one corner.

Step 1: Problem Understanding

Input

- Side of square
- Radius of quadrant

Output

Remaining area of square sheet

Formula

Area of square

= Side $\times$ Side

Area of quadrant

= $(\pi \times r^2)/4$

Remaining area

= Square Area – Quadrant Area

Step 2: Test Case Table

Test Case	Input (Side, Radius)	Expected Output	Actual Output	Result
1	10,5	80.37	Same	Pass
2	2,010	321.46	Same	Pass
3	15,3	217.93	Same	Pass

(Using $\pi = 3.14159$.)

Step 3: Algorithm

1. Start.
2. Read side of square.
3. Read radius.
4. Calculate area of square.
5. Calculate area of quadrant.
6. Calculate remaining area.
7. Display remaining area.
8. Stop.

Step 4: C Program

```
#include <stdio.h>

int main()
{
    float side, radius;

    float squareArea, quadrantArea, remainingArea;

    printf("Enter side of square: ");

    scanf("%f", &side);

    printf("Enter radius of quadrant: ");

    scanf("%f", &radius);

    squareArea = side * side;

    quadrantArea = (3.14159 * radius * radius) / 4;

    remainingArea = squareArea - quadrantArea;

    printf("\nRemaining Area = %.2f", remainingArea);

    return 0;
}
```

```

UN PICO 5.09                                     File: ques2.c
#include <stdio.h>
int main()
{
    float side, radius;
    float squareArea, quadrantArea, remainingArea;

    printf("Enter side of square: ");
    scanf("%f", &side);

    printf("Enter radius of quadrant: ");
    scanf("%f", &radius);

    squareArea = side * side;
    quadrantArea = (3.14159 * radius * radius) / 4;
    remainingArea = squareArea - quadrantArea;

    printf("\nRemaining Area = %.2f", remainingArea);

    return 0;
}

```

Step 5: OUTPUT

```

Last login: Sun Aug 30 22:34:08 on ttys000
aaryanrajp@earyans-macbook ~ % nano ques2.c
aaryanrajp@earyans-macbook ~ % nano ques2.c
aaryanrajp@earyans-macbook ~ % clang ques2.c -o ques2.c
aaryanrajp@earyans-macbook ~ % ./ques2.c
Enter side of square: 10
Enter radius of quadrant: 6

Remaining Area = 88.32
aaryanrajp@earyans-macbook ~ %

```